

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Midterm

Semester: Summer 2024

Duration: 80 minutes

Set - B

Full Marks: 40

CSE 420: Compiler Design

Figures in the right margin indicate marks.

Answer all the questions

<u>COs</u>	<u>Questions</u>	<u>Marks</u>
CO2	1. Construct the DFA from the following RE using Direct Method $(x \mid \epsilon)(x^* \mid y \mid z^* \mid \epsilon)^*(y \mid z)^*$ 	15
CO3	2. Consider the following <i>SLR Grammar</i> . Draw LR(0) automaton and construct SLR parse table 1. $V \rightarrow P Q$ 2. $P \rightarrow int$ 3. $P \rightarrow float$ 4. $Q \rightarrow \epsilon$ 5. $Q \rightarrow [size] Q$	10
CO3	3. Consider the SLR parse table from Question 2 . Show the parsing simulation using stack for the input string, <i>float [size]/[size]/[size]</i>	10
CO1	4. Explain the Errors in compiler	5

Augmentation:

$V' \rightarrow V$

$V \rightarrow PQ$

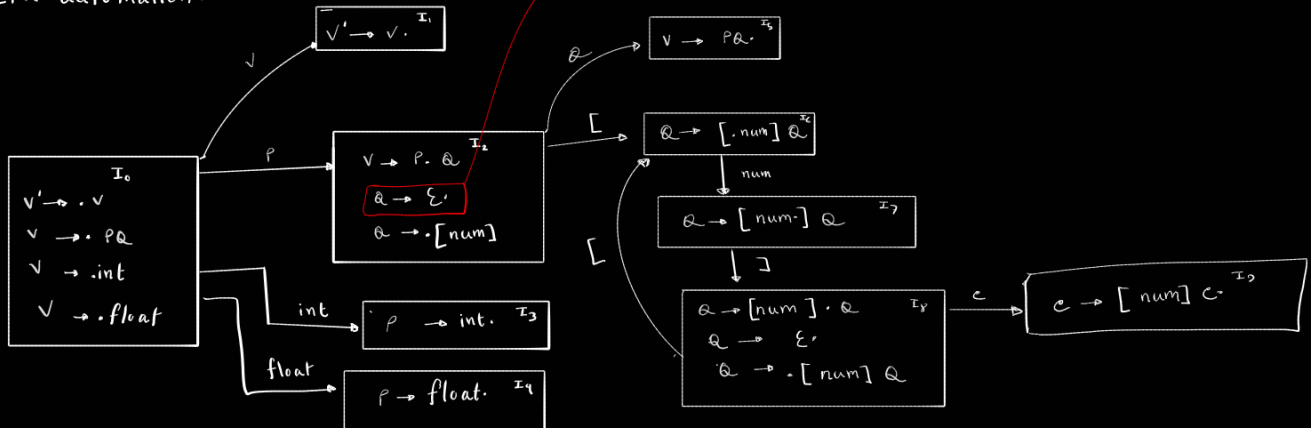
$P \rightarrow \text{int}$

$P \rightarrow \text{float}$

$Q \rightarrow \epsilon$

$Q \rightarrow [\text{num}] Q$

LR(0) automation:



$\text{First}(V) = \{\text{int}, \text{float}\}$

$\text{First}(P) = \{\text{int}, \text{float}\}$

$\text{First}(Q) = \{[, \epsilon]\}$

$\text{Follow}(V) = \{\#\}$

$\text{Follow}(P) = \{[, \#\}$

$\text{Follow}(Q) = \{\#\}$

$V \rightarrow PQ$ (1)

$P \rightarrow \text{int}$ (2)

$P \rightarrow \text{float}$ (3)

$Q \rightarrow \epsilon$ (4)

$Q \rightarrow [\text{num}] Q$ (5)

state	Action						Go-to		
	[	num	]	int	float	#	V	P	Q
0				S3	S4		1	2	
1						Accept			
2	S6					R4			5
3	R2					R2			
4	R3					R3			
5						R1			
6		S7							
7				S8					
8	S6					R4			9
9						R5			

Stack	Symbol	Input buffer	Action
0	#	int [num] [num] [num] #	S
0 4	# float	[num] [num] [num] #	Reduce by $P \rightarrow \text{float}$.
0 2	# P	[num] [num] [num] #	S6
0 2 6	# P [	num] [num] [num] #	S7
0 2 6 7	# P [num	] [num] [num] #	S8
0 2 6 7 8	# P [num]	[num] [num] #	S6
0 2 6 7 8 6	# P [num] [	num] [num] #	S7
0 2 6 7 8 6 7	# P [num] [num	] [num] #	S8
0 2 6 7 8 6 7 8	# P [num] [num]	[num] #	S6
0 2 6 7 8 6 7 8 6	# P [num] [num] [	num] #	S7
0 2 6 7 8 6 7 8 6 7	# P [num] [num] [num	] #	S8
0 2 6 7 8 6 7 8 6 7 8	# P [num] [num] [num]	#	Reduce by $Q \rightarrow \epsilon$
0 2 6 7 8 6 7 8 6 7 8 9	# P [num] [num] [num] Q	#	Reduce by $Q \rightarrow [\text{num}] Q$
0 2 6 7 8 6 7 8 6 7 8 9 9	# P [num] [num] [num] Q	#	Reduce by $Q \rightarrow [\text{num}] Q$
0 2 6 7 8 6 7 8 6 7 8 9 9 9	# P [num] [num] [num] Q	#	Reduce by $Q \rightarrow [\text{num}] Q$
0 2 6 7 8 6 7 8 6 7 8 9 9 9 9	# P [num] [num] [num] Q	#	Reduce by $V \rightarrow PQ$
0 1	# V	#	Accept